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Inspection apparatus and inspection system

Abstract

An inspection apparatus of an embodiment includes a rotor image-pickup unit for picking up an image of a rotor using a rotor image-pickup device in the air gap between the rotor and a stator. The rotor image-pickup device includes: a carriage casing; a camera; an image-pickup position changing part; and an image-pickup direction changing part. The camera is installed in the carriage casing, and picks up an image of a ventilation hole of the rotor in the air gap. The image-pickup position changing part is installed in the carriage casing, and changes an image-pickup position of the camera by moving the carriage casing in a circumferential direction of the rotor in the air gap. The image-pickup direction changing part is installed in the carriage casing, and changes an image-pickup direction of the camera to an inclined angle to a radial direction of the rotor in the air gap.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS

(1) This application is based upon and claims the benefit of priority from Japanese Patent Application No. 2022-158625, filed on Sep. 30, 2022, and Japanese Patent Application No. 2022-158633, filed on Sep. 30, 2022; the entire contents of which are incorporated herein by reference.

FIELD

(2) Embodiments relate to an inspection apparatus and an inspection system.

BACKGROUND

(3) In rotary electric machines such as a generator and an electric motor, a cylindrical gap is interposed between a stator and a rotor.

(4) On the rotary electric machine, an inspection for electrical and mechanical soundness is performed by maintenance work. The inspection on the rotary electric machine is performed by pulling the rotor out of the stator, for example. In addition to this, the inspection on the rotary electric machine is performed by inserting an inspection apparatus into a gap between the stator and the rotor and moving it in the gap, for example.

(5) However, in the conventional inspection apparatus, the inspection is difficult to perform efficiently in some cases because the inspection apparatus is not easy to insert into the gap between the stator and the rotor, or the like. In particular, when the inspection is performed on the rotary electric machine in which a ventilation hole for cooling is formed in the rotor, in a case of inserting the inspection apparatus into the gap and making the inspection apparatus adhere to the stator, the rotor is in a state of stopping in an arbitrary position, which does not facilitate sufficient observation to a deep portion of the ventilation hole. As a result, it may be difficult to determine whether or not the ventilation hole is closed.

(6) Accordingly, a problem to be solved by the present invention is to provide an inspection apparatus and an inspection system which easily enable an efficient inspection.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a view schematically illustrating one example of a rotary electric machine **10** which is an inspection object in an inspection apparatus of an embodiment.

(2) FIG. 2 is a block diagram schematically illustrating an inspection system **800** for inspecting the rotary electric machine **10** (refer to FIG. 1) in the embodiment.

(3) FIG. 3A is a perspective view illustrating an inspection apparatus **500** in the embodiment.

(4) FIG. 3B is a perspective view illustrating the inspection apparatus **500** in the embodiment.

(5) FIG. 3C is a perspective view illustrating a state of disassembling a part of the inspection apparatus **500** of the embodiment.

(6) FIG. 4A is a view for explaining portions where a plurality of inspection units **511** to **517** are coupled to one another in the inspection apparatus **500** of the embodiment.

(7) FIG. 4B is a view for explaining the portions where the plurality of inspection units **511** to **517** are coupled to one another in the inspection apparatus **500** of the embodiment.

(8) FIG. 5 is a perspective view illustrating the other example of a coupled state in which the plurality of inspection units **511** to **517** are coupled to one another in the inspection apparatus **500** of the embodiment.

- (9) FIG. 6A is a perspective view illustrating the inspection unit **513** in the inspection apparatus **500** of the embodiment (assembled state).
- (10) FIG. 6B is a perspective view illustrating the inspection unit **513** in the inspection apparatus **500** of the embodiment (disassembled state).
- (11) FIG. 6C is a perspective view illustrating the inspection unit **513** in the inspection apparatus **500** of the embodiment (open state).
- (12) FIG. 7A is a perspective view illustrating an image-pickup device **631** in the inspection unit **513** of the embodiment.
- (13) FIG. 7B is a top view illustrating the image-pickup device **631** in the inspection unit **513** of the embodiment.
- (14) FIG. 7C is an exploded perspective view illustrating a state of disassembling the image-pickup device **631** in the inspection unit **513** of the embodiment.
- (15) FIG. 8 is a bottom view of a camera **910** in the image-pickup device **631** of the embodiment.
- (16) FIG. 9A is a side view illustrating an image-pickup direction changing part **950** of the image-pickup device **631** in the inspection unit **513** of the embodiment.
- (17) FIG. 9B is a side view illustrating the image-pickup direction changing part **950** of the image-pickup device **631** in the inspection unit **513** of the embodiment.
- (18) FIG. 9C is a side view illustrating the image-pickup direction changing part **950** of the image-pickup device **631** in the inspection unit **513** of the embodiment.
- (19) FIG. 10A is a view illustrating a state of the inspection apparatus **500** when the inspection is performed on the rotary electric machine **10** in the embodiment.
- (20) FIG. 10B is a view illustrating a state of the inspection apparatus **500** when the inspection is performed on a ventilation hole of a rotor **20** constituting the rotary electric machine **10** in the embodiment.
- (21) FIG. 11 is a perspective view illustrating the inspection unit **513** in a modified example of the embodiment.

DETAILED DESCRIPTION

(22) An inspection apparatus of an embodiment, when an inspection is performed on a rotary electric machine in which an air gap is interposed between a rotor and a stator, and a ventilation hole along a radial direction of the rotor is formed in the rotor, is inserted into the air gap, and adheres to the stator. The inspection apparatus of the embodiment includes a rotor image-pickup unit for picking up an image of the rotor using a rotor image-pickup device in the air gap. The rotor image-pickup device includes: a carriage casing; a camera; an image-pickup position changing part; and an image-pickup direction changing part. The camera is installed in the carriage casing, and picks up an image of the ventilation hole of the rotor in the air gap. The image-pickup position changing part is installed in the carriage casing, and configured to change an image-pickup position of the camera by moving the carriage casing in a circumferential direction of the rotor in the air gap. The image-pickup direction changing part is installed in the carriage casing, and configured to change an image-pickup direction of the camera to an inclined angle to the radial direction of the rotor in the air gap.

(23) One example of a rotary electric machine **10** which is an inspection object will be described before describing an inspection apparatus and an inspection system of an embodiment.

[A] Rotary Electric Machine **10**

(24) FIG. 1 is a view schematically illustrating one example of the rotary electric machine **10** which is the inspection object in the inspection apparatus and the inspection system of the embodiment.

(25) As illustrated in FIG. 1, the rotary electric machine **10**, in which a rotor **20** and a stator **40** are housed in a rotary electric machine casing **60**, is an inner rotor type of the rotor **20** rotating inside the stator **40**. The rotary electric machine **10** is, for example, a generator, and is configured to generate electricity by rotation of the rotor **20**. The rotary electric machine **10** can be used as an

electric motor in addition to the generator.

(26) In this embodiment, the rotary electric machine **10** is configured such that a cooling gas CG (a cooling medium such as hydrogen or air) flows inside the rotary electric machine casing **60**, thereby cooling the rotor **20** and the stator **40**.

(27) The components of the rotary electric machine **10** will be described.

(28) [A-1] Rotor **20**

(29) In the rotary electric machine **10**, the rotor **20** is installed so that an axial direction along a rotation axis AX is along a horizontal direction x. Here, in the rotor **20**, a rotor core **200** in a cylindrical shape is provided coaxially with a rotary shaft **201**, and a cooling flow path through which the cooling gas CG flows is formed.

(30) In the rotor core **200** of the rotor **20**, a plurality of rotor slots (illustration is omitted) are formed in a circumferential direction of the rotor **20**, and in the rotor slot, members of a rotor coil, a rotor wedge, and the like are installed. In the members installed in the rotor slot, a ventilation hole through which the cooling gas CG flows is formed along a radial direction of the rotor **20**. A plurality of the ventilation holes through which the cooling gas CG flows are formed at intervals in the circumferential direction of the rotor **20**, and the plurality of ventilation holes are formed at intervals in the axial direction of the rotor **20**.

(31) [A-2] Stator **40**

(32) In the stator **40**, a stator coil **42** is installed in a stator core **41**. The stator core **41** is in a cylindrical shape, and is provided coaxially with the rotary shaft **201**. The stator core **41** is installed to surround the rotor core **200** with an air gap AG in a cylindrical shape therebetween.

(33) On an inner peripheral side of the stator core **41**, stator slots penetrating in the axial direction of the rotation axis AX are formed, and the stator coil **42** is inserted into the stator slot, and fixed with, for example, a stator wedge (illustration is omitted). Although illustration is omitted, a cooling flow path through which the cooling gas CG flows is also formed in the stator **40**.

(34) [A-3] Rotary Electric Machine Casing **60**

(35) In the rotary electric machine casing **60**, which has a double structure, an outer casing **62** is provided outside an inner casing **61**. The inner casing **61** has a through hole K**61** penetrated by the rotary shaft **201**, and houses the rotor **20** and the stator **40** in the interior thereof. The outer casing **62** has a through hole K**62** penetrated by the rotary shaft **201**, and houses the inner casing **61** in the interior thereof.

(36) In the rotary electric machine casing **60**, an opening K**611** is provided on an upper side of the inner casing **61**, and a gas cooler **80** is attached to the opening K**611**. Further, a fan **90** is housed inside the through hole K**61** of the inner casing **61**.

(37) The fan **90** is an axial fan, and is fixed to each of one side portion and the other side portion of the rotary shaft **201** so as to sandwich the rotor **20**.

(38) [A-4] Flow of the Cooling Gas CG in the Rotary Electric Machine **10**

(39) In the rotary electric machine **10**, by rotation of the fan **90** with the rotary shaft **201**, the cooling gas CG flows from the exterior of the inner casing **61** into the interior thereof. Here, the cooling gas CG flows to go from each of one side portion and the other side portion of the rotary shaft **201** toward the middle portion thereof.

(40) Inside the inner casing **61**, the cooling gas CG flows into the cooling flow path formed inside the rotor **20**, thereafter flowing out to the air gap AG located between an outer peripheral surface of the rotor **20** and an inner peripheral surface of the stator **40**. In the rotor **20**, the cooling gas CG flows along the axial direction, thereafter flowing from its inner peripheral side to its outer peripheral side through the ventilation holes formed along the radial direction.

(41) Although illustration is omitted, the cooling gas CG thereafter flows via the air gap AG through the cooling flow path formed inside the stator **40**. The cooling gas CG discharged to the exterior of the stator **40** flows from the interior of the inner casing **61** to the exterior thereof via the gas cooler **80**. At this time, the cooling gas CG is cooled in the gas cooler **80**. The cooling gas CG

cooled in the gas cooler **80** flows through a space located outside the inner casing **61** in the outer casing **62**, thereafter, as described above, flowing from the exterior of the inner casing **61** into the interior thereof by the rotation of the fan **90**.

(42) Thus, the rotary electric machine **10** is configured to cool the respective components by the cooling gas CG flowing in circulation inside the rotary electric machine casing **60**.

[B] Inspection System **800**

(43) An inspection system **800** used when the above-described rotary electric machine **10** (refer to FIG. **1**) is inspected will be described.

(44) FIG. **2** is a block diagram schematically illustrating the inspection system **800** for inspecting the rotary electric machine **10** (refer to FIG. **1**) in the embodiment.

(45) As illustrated in FIG. **2**, the inspection system **800** of the embodiment includes an inspection apparatus **500** and a control apparatus **600**, and is configured to be capable of communicating between the inspection apparatus **500** and the control apparatus **600** via a cable **700**.

(46) [B-1] Inspection Apparatus **500**

(47) Although details are described later, the inspection apparatus **500** is a self-propelled inspection robot, and is configured to, when the inspection on the rotary electric machine **10** (refer to FIG. **1**) is performed, be inserted into the air gap AG interposed between the rotor **20** and the stator **40** and be capable of traveling in a state of adhering to the stator **40**.

(48) [B-2] Control Apparatus **600**

(49) The control apparatus **600** is provided for remotely controlling the operation of the inspection apparatus **500**. The control apparatus **600** includes an arithmetic unit (computer) and a storage device, and is configured such that the arithmetic unit executes the control of the operation of the inspection apparatus **500** using programs stored by the storage device.

(50) Here, the control apparatus **600** performs the control so that the inspection apparatus **500** performs the inspection on the rotary electric machine **10** in response to an operation command input with operating devices (mouse, keyboard, and the like) by an operator, for example. Further, the control apparatus **600** performs the control so as to display information on the inspection to be performed by the inspection apparatus **500**, information obtained by performing the inspection with the inspection apparatus **500**, and the like on, for example, a display.

[C] Configuration of the Inspection Apparatus **500**

(51) A configuration of the inspection apparatus **500** constituting the above-described inspection system **800** (refer to FIG. **2**) will be described.

(52) FIG. **3A** and FIG. **3B** are each a perspective view illustrating the inspection apparatus **500** in the embodiment.

(53) When the inspection apparatus **500** is inserted into the air gap AG between the rotor **20** and the stator **40**, FIG. **3A** illustrates its surface side located on the stator **40** side. When the inspection apparatus **500** is inserted into the air gap AG between the rotor **20** and the stator **40**, FIG. **3B** illustrates its surface side located on the rotor **20** side.

(54) As illustrated in FIG. **3A** and FIG. **3B**, the inspection apparatus **500** includes a plurality of inspection units **511** to **517** and a coupling shaft **520**.

(55) [C-1] a Plurality of Inspection Units **511** to **517**

(56) A plurality of the respective inspection units **511** to **517** are provided for performing the inspection in the air gap AG between the rotor **20** and the stator **40** constituting the rotary electric machine **10** (refer to FIG. **1**). A plurality of the respective inspection units **511** to **517** are configured to be in a state of being coupled in an axial direction along the rotation axis AX of the rotor **20** when inserted into the air gap AG between the rotor **20** and the stator **40**.

(57) A plurality of the respective inspection units **511** to **517** have functions for performing the inspection on the rotary electric machine **10** (refer to FIG. **1**).

(58) Concretely, the inspection unit **511** is a tip-side image-pickup unit, includes an image-pickup device **611** and a lighting device **612** as inspection devices, and is configured such that the lighting

device **612** lights a tip side of the inspection apparatus **500** and the image-pickup device **611** picks up its image (refer to FIG. **3A**).

(59) The inspection unit **512** is a guide unit, includes a pair of guide plates **621** as an inspection device, and is configured to guide a moving direction of the inspection apparatus **500**. Here, a pair of the guide plates **621** is inserted into a groove (illustration is omitted) formed along the axial direction in the stator **40** (refer to FIG. **1**), and guides the inspection apparatus **500** in moving along the axial direction, for example. Further, the inspection unit **512** is configured to vary a distance between a pair of the guide plates **621** according to a width of the groove formed in the stator **40** (refer to FIG. **3A**).

(60) The inspection unit **513** is a rotor image-pickup unit, includes an image-pickup device **631** (rotor image-pickup device) as an inspection device, and is configured to pick up an image of the rotor **20** (refer to FIG. **1**) with the image-pickup device **631** (refer to FIG. **3B**).

(61) The inspection unit **514** is a traveling unit, and includes a moving mechanism including a crawler **641** (endless belt) and a permanent magnet **642** as an inspection device. The inspection unit **514** is configured such that the inspection apparatus **500** moves with the crawler **641** in a state in which the inspection apparatus **500** adheres to the stator **40** (refer to FIG. **1**) by magnetic force of the permanent magnet **642**. Here, the inspection unit **514** includes the two moving mechanisms in each of which the one permanent magnet **642** is sandwiched by a pair of the crawlers **641**, and the operation of the two moving mechanisms allows the inspection apparatus **500** to be moved to any position. Further, the inspection unit **514** includes a hammer **643** and a microphone **644** as inspection devices, and is configured to detect sound when the stator **40** is struck with the hammer **643** by using the microphone **644**. In addition to this, the inspection unit **514** includes an image-pickup device **645** as an inspection device, and is configured to pick up an image of the stator **40** with the image-pickup device **645** (refer to FIG. **3A**).

(62) The inspection unit **515** is a control unit, and is configured to include a control board (illustration is omitted) for controlling the operation of the units constituting the inspection apparatus **500** as an inspection device.

(63) The inspection unit **516** is a guide unit, includes a pair of guide plates **661** as an inspection device, and is configured to guide the moving direction of the inspection apparatus **500**, similarly to the inspection unit **512** (refer to FIG. **3A**).

(64) The inspection unit **517** is an interface to which the cable **700** for connecting the inspection apparatus **500** and the control apparatus **600** (refer to FIG. **2**) therebetween is coupled. Further, the inspection unit **517** includes an image-pickup device **671** as an inspection device, and is configured such that the image-pickup device **671** picks up an image on a rear end side of the inspection apparatus **500** (refer to FIG. **3A**).

(65) FIG. **3C** is a perspective view illustrating a state of disassembling a part of the inspection apparatus **500** of the embodiment.

(66) As illustrated in FIG. **3C**, through each of the inspection units **511** to **514**, a coupling hole **H510** is formed. Although illustration is omitted, the coupling hole **H510** is also formed through each of the other inspection units **515** to **517**. In the plurality of inspection units **511** to **517**, the coupling holes **H510** are configured to be communicated with one another when the plurality of inspection units **511** to **517** are in a coupled state.

(67) In this embodiment, the two coupling holes **H510** are formed through each of the plurality of inspection units **511** to **517**. The two coupling holes **H510** are provided at both ends of each of the plurality of inspection units **511** to **517**.

(68) Note that although illustration is omitted, the plurality of inspection units **511** to **517** each have a connector (plug, jack, and the like) capable of electrically connecting one another, and are configured to be capable of communicating with the control apparatus **600**.

(69) [C-2] Coupling Shaft **520**

(70) The coupling shaft **520** is a rod-shaped body, and is provided for coupling the plurality of

inspection units **511** to **517** therebetween, as illustrated in FIG. 3C.

(71) Here, the coupling shaft **520** is detachable with respect to the plurality of inspection units **511** to **517**. The coupling shafts **520** are inserted into the coupling holes **H510** communicated with one another in the plurality of inspection units **511** to **517** when the plurality of inspection units **511** to **517** falls into the coupled state. In contrast with this, when the coupled state of the plurality of inspection units **511** to **517** is released, the coupling shafts **520** are taken out of the coupling holes **H510**.

(72) [C-3] Details of Coupling Portions of the Plurality of Inspection Units **511** to **517**

(73) Details of portions where the plurality of inspection units **511** to **517** are coupled to one another will be described.

(74) FIG. 4A and FIG. 4B are each a view for explaining the portions where the plurality of inspection units **511** to **517** are coupled to one another in the inspection apparatus **500** of the embodiment.

(75) FIG. 4A is a perspective view of enlarging and illustrating the coupling portions of the inspection units **511** to **514** in the inspection apparatus **500**. FIG. 4B is a plan view of enlarging and illustrating the coupling portions of the inspection units **511** to **514** in the inspection apparatus **500**. Here, the coupling portions of the inspection units **511** to **514** are illustrated, and they are also the same with coupling portions of the other inspection units **515** to **517**.

(76) As illustrated in FIG. 4A and FIG. 4B, the inspection units **511** to **514** are configured to be coupled by being fitted into one another. Concretely, a male fitting portion **511M** formed in the inspection unit **511** and a female fitting portion **512F** formed in the inspection unit **512** are fitted into each other. Further, a male fitting portion **512M** formed in the inspection unit **512** and a female fitting portion **513F** formed in the inspection unit **513** are fitted into each other. Similarly, a male fitting portion **513M** formed in the inspection unit **513** and a female fitting portion **514F** formed in the inspection unit **514** are fitted into each other.

(77) Although illustration is omitted, any of the male fitting portions **511M**, **512M**, **513M** and any of the female fitting portions **512F**, **513F**, **514F** are configured to be fitted into each other. Thus, the plurality of inspection units **511** to **517** are detachable between one another, which allows various coupled states in addition to the above-described coupled state.

(78) FIG. 5 is a perspective view illustrating the other example of a coupled state in which the plurality of inspection units **511** to **517** are coupled to one another in the inspection apparatus **500** of the embodiment.

(79) As illustrated in FIG. 5, in the inspection apparatus **500** of the embodiment, differently from the case in FIG. 3A, for example, the inspection unit **512** and the inspection unit **514** can be coupled to each other without interposing the inspection unit **513** therebetween. Thus, in this embodiment, the plurality of inspection units **511** to **517** have a structure common to their fitting portions, which allows any change in a length of the inspection apparatus **500**.

(80) Incidentally, in this case, the coupling shaft **520** having a length corresponding to the length of the inspection apparatus **500** is used.

[D] Detailed Configuration of the Inspection Unit **513** (Rotor Image-Pickup Unit)

(81) Of the plurality of inspection units **511** to **517**, the configuration of the inspection unit **513** which is the rotor image-pickup unit will be described.

(82) FIG. 6A, FIG. 6B, and FIG. 6C are each a perspective view illustrating the inspection unit **513** (rotor image-pickup unit) in the inspection apparatus **500** of the embodiment.

(83) FIG. 6A illustrates a state of assembling the inspection unit **513** (rotor image-pickup unit).

FIG. 6B illustrates a state of disassembling a part of the inspection unit **513** (rotor image-pickup unit). Further, FIG. 6C illustrates a state of opening a part of the inspection unit **513** (rotor image-pickup unit) (here, a state of removing a part of a first frame member **630a**) for explanation.

(84) [D-1] Frame **630**

(85) As illustrated in FIG. 6A to FIG. 6C, the inspection unit **513** includes a frame **630** housing the

image-pickup device **631** in addition to the image-pickup device **631** which is the rotor image-pickup device (refer to FIG. 3B).

(86) In the inspection unit **513**, as illustrated in FIG. 6A, the frame **630** is in an arc shape along the air gap AG in the cylindrical shape (refer to FIG. 1). In this embodiment, in the frame **630**, at least a surface is formed using resin. For example, the whole of the frame **630** is formed using nylon 12. In addition to this, the frame **630** may be formed by covering a surface of a base material formed of a metallic material with the resin, for example. Similarly, the inspection units **511**, **512**, **514** to **517** in addition to the inspection unit **513** each also include an arc-shaped frame, and the frame is formed using the resin.

(87) The frame **630** includes the first frame member **630a** and a second frame member **630b**, and the first frame member **630a** and the second frame member **630b** are, for example, fitted into each other, as illustrated in FIG. 6B. Note that similarly, the inspection units **511**, **512**, **514** to **517** in addition to the inspection unit **513** each also include the arc-shaped frame.

(88) As illustrated in FIG. 6C, a rack gear **801** is formed in the inner portion of the first frame member **630a** constituting the frame **630**. The rack gear **801** is configured to arrange a plurality of gear teeth side by side along the arc shape of the frame **630**.

(89) [D-2] Image-Pickup Device **631**

(90) In the inspection unit **513**, the image-pickup device **631** is configured to be capable of varying (refer to an arrow AR1) an image-pickup position, and be capable of varying (refer to an arrow AR2) an image-pickup direction by moving along the arc shape of the frame **630** inside the frame **630**, as illustrated in FIG. 6A.

(91) As illustrated in FIG. 6B, the image-pickup device **631** is electrically connected to a flexible cable **632** inside the frame **630**. The image-pickup device **631** is electrically connected via the flexible cable **632** to the control apparatus **600** (refer to FIG. 2), and its operation is controlled.

(92) FIG. 7A is a perspective view illustrating the image-pickup device **631** in the inspection unit **513** of the embodiment. FIG. 7B is a top view illustrating the image-pickup device **631** in the inspection unit **513** of the embodiment. FIG. 7C is an exploded perspective view illustrating a state of disassembling the image-pickup device **631** in the inspection unit **513** of the embodiment.

(93) In the inspection unit **513**, a camera **910**, an image-pickup position changing part **930**, and an image-pickup direction changing part **950** are installed in a carriage casing **900** in the image-pickup device **631** which is the rotor image-pickup device, as illustrated in FIG. 7A to FIG. 7C.

(94) Components of the image-pickup device **631** will be described in order.

(95) [D-2-1] Carriage Casing **900**

(96) In the image-pickup device **631**, the carriage casing **900**, whose outer shape is a rectangular parallelepiped shape, includes a housing space SP910 housing the camera **910**, a housing space SP930 housing the image-pickup position changing part **930**, and a housing space SP950 housing the image-pickup direction changing part **950**. The carriage casing **900** is configured to sandwich the housing space SP910 between the housing space SP930 and the housing space SP950 (refer to FIG. 7B, FIG. 7C).

(97) [D-2-2] Camera **910**

(98) In the image-pickup device **631**, the camera **910**, whose details are described later, is provided for picking up an image of the ventilation hole in the air gap AG between the rotor **20** and the stator **40** (refer to FIG. 1).

(99) Here, the camera **910** is installed in the housing space SP910 of the carriage casing **900** in a state of being housed in a camera holder **911** (refer to FIG. 7A to FIG. 7C).

(100) The camera holder **911** is in a cylindrical shape, and houses the camera **910** in the interior thereof. Further, on an outer peripheral surface of the camera holder **911**, a spur gear **912** is provided. The spur gear **912** is provided so that a rotation axis of the spur gear **912** is along a normal direction of the camera holder **911** in the cylindrical shape. In addition to this, on the outer peripheral surface of the camera holder **911**, a shaft pin **911a** is provided. The shaft pin **911a** is in a

column shape, and is provided so that a center axis of the shaft pin **911a** is aligned coaxially with the rotation axis of the spur gear **912** (refer to FIG. 7A to FIG. 7C).

(101) Further, the camera holder **911** is installed in the carriage casing **900** so as to swing along a pivoting direction of the spur gear **912** using a bush **913** and a bush **914** (refer to an arrow AR2 in FIG. 7A).

(102) The bush **913** is coupled to the spur gear **912** with a hexagon socket set screw HS therebetween. Further, the bush **913** is fixed to the carriage casing **900** using a screw SC1. The bush **914** is coupled to the shaft pin **911a**. Further, the bush **914** is fixed in the carriage casing **900** using a screw SC2 (refer to FIG. 7C).

(103) FIG. 8 is a bottom view of the camera **910** in the image-pickup device **631** of the embodiment.

(104) As illustrated in FIG. 8, the camera **910** is provided with a lighting section **910b** so as to surround a peripheral surface of an image-pickup lens **910a** in a disc shape. The lighting section **910b** is constituted using, for example, a light-emitting diode (LED).

(105) [D-2-3] Image-Pickup Position Changing Part **930**

(106) The image-pickup position changing part **930** is provided for changing an image-pickup position of the camera **910** by moving the carriage casing **900** in the circumferential direction of the rotor **20** (refer to the arrows AR1 in the figures) in the air gap AG between the rotor **20** and the stator **40** (refer to FIG. 1).

(107) Here, the image-pickup position changing part **930** is installed in the housing space SP**930** of the carriage casing **900**, and includes an image-pickup position changing motor **931** and an image-pickup position changing pinion gear **932** (refer to FIG. 7A to FIG. 7C).

(108) In the image-pickup position changing part **930**, the image-pickup position changing motor **931** is fixed in the carriage casing **900** using a retaining plate **933** and a screw SC3. The image-pickup position changing pinion gear **932** is installed on a rotation shaft of the image-pickup position changing motor **931** (refer to FIG. 7A to FIG. 7C).

(109) As illustrated in FIG. 6C, in the inspection unit **513**, the image-pickup device **631** is installed so that the image-pickup position changing pinion gear **932** engages with the rack gear **801** formed in the frame **630**. Thus, in this embodiment, a rotation of the image-pickup position changing pinion gear **932** moves the image-pickup device **631** along a drawing direction of the rack gear **801**.

(110) [D-2-4] Image-Pickup Direction Changing Part **950**

(111) The image-pickup direction changing part **950** is provided for changing an image-pickup direction of the camera **910** to an inclined angle to the radial direction of the rotor **20** (dotted lines in FIG. 6A, FIG. 6C, and FIG. 7A) in the air gap AG between the rotor **20** and the stator **40** (refer to FIG. 1) (refer to the arrows AR2 in FIG. 6A, FIG. 6C, and FIG. 7A).

(112) Here, the image-pickup direction changing part **950** is installed in the housing space SP**950** of the carriage casing **900**, and includes an image-pickup direction changing motor **951** and an image-pickup direction changing pinion gear **952** (refer to FIG. 7A to FIG. 7C).

(113) In the image-pickup direction changing part **950**, the image-pickup direction changing motor **951** is fixed in the carriage casing **900** using a retaining plate **953** and a screw SC4. The image-pickup direction changing motor **951** is housed in the carriage casing **900** so that its rotation shaft is parallel to the rotation shaft of the image-pickup position changing motor **931**. Further, the image-pickup direction changing pinion gear **952** is installed on the rotation shaft of the image-pickup direction changing motor **951** (refer to FIG. 7A to FIG. 7C).

(114) FIG. 9A to FIG. 9C are each a side view illustrating the image-pickup direction changing part **950** of the image-pickup device **631** in the inspection unit **513** of the embodiment.

(115) FIG. 9A to FIG. 9C each illustrate a state of the image-pickup device **631** in a vertical plane (yz plane) orthogonal to the axial direction (x direction) when the inspection apparatus **500** is inserted into the air gap AG between the rotor **20** and the stator **40** (refer to FIG. 1) in the rotary

electric machine **10** (refer to FIG. **1**). FIG. **9A** illustrates the state in which the image-pickup direction of the camera **910** is along the radial direction of the rotor **20**. In contrast with this, FIG. **9B** and FIG. **9C** each illustrate the state in which the image-pickup direction of the camera **910** is along the radial direction of the rotor **20**. In FIG. **9A** to FIG. **9C**, for convenience of explanation, illustration of members except for parts related to the image-pickup direction changing part **950** is appropriately omitted.

(116) As illustrated in FIG. **9A** to FIG. **9C**, the image-pickup direction changing part **950** is installed in the carriage casing **900** so that the image-pickup direction changing pinion gear **952** engages with the spur gear **912** provided on the camera holder **911**. Thus, in the image-pickup device **631** of this embodiment, a rotation of the image-pickup direction changing pinion gear **952** changes the image-pickup direction of the camera **910** housed in the camera holder **911** from the state of being along the radial direction of the rotor **20** to the state of being inclined thereto.

(117) [D-2-5] Others

(118) In addition to the above, the image-pickup device **631** includes a cable cover **960**, pins **970**, and a cable retaining plate **980** as illustrated in FIG. **7A** to FIG. **7C**.

(119) The cable cover **960** is fixed in the carriage casing **900** using a screw SC5 (refer to FIG. **7C**).

(120) A plurality of the pins **970** are installed on the carriage casing **900** as illustrated in FIG. **7A** to FIG. **7C**. Here, in the carriage casing **900**, on each of both side surfaces along the direction in which the image-pickup direction changing part **950** moves the carriage casing **900** (the arrow AR1), a pair of insertion holes H970 is formed at an interval, and the pins **970** are inserted into the respective insertion holes H970.

(121) The cable retaining plate **980** is fixed to the carriage casing **900** using a screw SC6 (refer to FIG. **7C**).

[E] Inspection Method

(122) States when the inspection on the rotary electric machine **10** (refer to FIG. **1**) is performed using the above-described inspection apparatus **500** (refer to FIG. **3A**) will be described.

(123) FIG. **10A** is a view illustrating the state of the inspection apparatus **500** when the inspection is performed on the rotary electric machine **10** in the embodiment. FIG. **10A** illustrates the vertical plane (yz plane) orthogonal to the axial direction (x direction) in the rotary electric machine **10** (refer to FIG. **1**).

(124) As illustrated in FIG. **10A**, when the inspection on the rotary electric machine **10** is performed, the inspection apparatus **500** is inserted into the air gap AG interposed between the rotor **20** and stator **40**. Here, the inspection apparatus **500** is inserted into the air gap AG of the rotary electric machine **10** so that a coupling direction of the plurality of inspection units **511** to **517** constituting the inspection apparatus **500** (refer to FIG. **3A**) is along the axial direction of the rotary electric machine **10** (in FIG. **10A**, the x direction).

(125) The inspection apparatus **500** inserted into the air gap AG travels using the crawler **641** in the state of adhering to the stator **40** by magnetic force of the permanent magnet **642**. Here, the inspection apparatus **500** travels in a state in which the crawler **641** and the stator **40** are in close contact with each other, in contrast with a state in which the permanent magnet **642** and the stator **40** are not in close contact with but separate from each other.

(126) Although illustration is omitted in FIG. **10A**, in the inspection apparatus **500**, the guide plates **621** and the guide plates **661** (refer to FIG. **3A**) are inserted into the groove formed along the axial direction in the stator **40**. Thus, the inspection apparatus **500** travels along the axial direction with the guide plates **621** and the guide plates **661** (refer to FIG. **3A**). An operator observes images picked up by the image-pickup device **611**, the image-pickup device **631**, the image-pickup device **645**, and the image-pickup device **671** provided in the inspection apparatus **500**, and the operator operates the inspection apparatus **500**, thereby performing the movement of the inspection apparatus **500** (refer to FIG. **3A**, FIG. **3B**).

(127) Then, the inspection on the rotary electric machine **10** is performed using the functions

included in the inspection apparatus **500** (refer to FIG. 3A).

(128) In this embodiment, by observing digital moving images obtained by picking up images of the inner peripheral surface of the stator **40** with the image-pickup device **645** in real time, a condition of the stator **40** is inspected. In addition to this, to inspect looseness of the stator wedge (illustrated is omitted) for fixing the stator coil **42** to the stator core **41** in the stator **40**, the inspection in which the stator wedge is struck with the hammer **643** to detect the sound at the time of striking with the microphone **644** is performed.

(129) Further, in this embodiment, by observing digital moving images obtained by picking up images of the outer peripheral surface of the rotor **20** with the image-pickup device **631** in real time, a condition of the rotor **20** is inspected. Here, the inspection in which an image of the ventilation hole extending in the radial direction in the rotor **20** is picked up and the ventilation hole is whether to be in a closed state is performed.

(130) FIG. **10B** is a view illustrating the state of the inspection apparatus **500** when the inspection is performed on the ventilation hole of the rotor **20** constituting the rotary electric machine **10** in the embodiment. FIG. **10B** illustrates the vertical plane (yz plane) orthogonal to the axial direction (x direction) similarly to FIG. **10A**.

(131) As illustrated in FIG. **10B**, when the inspection is performed on a ventilation hole V of the rotor **20** constituting the rotary electric machine **10**, the inspection unit **513** constituting the rotor image-pickup unit in the inspection apparatus **500** is used. At this time, in the inspection unit **513**, to make the image-pickup position of the camera **910** correspond to the ventilation hole V, the image-pickup position changing part **930** moves the image-pickup device **631** in the circumferential direction of the rotor **20** (the arrow AR1). Further, in the inspection unit **513**, the image-pickup direction changing part **950** changes the image-pickup direction of the camera **910** (the arrow AR2). In this embodiment, this causes the image-pickup direction of the camera **910** to be along a depth direction of the ventilation hole V, which allows an image of the ventilation hole V to be picked up from directly thereabove.

[F] Summary

(132) As described above, in the inspection apparatus **500** of this embodiment, the inspection unit **513** which is the rotor image-pickup unit includes the rotor image-pickup device **631** including the image-pickup position changing part **930** and the image-pickup direction changing part **950**. The image-pickup position changing part **930** is configured to change the image-pickup position of the camera **910** by moving the carriage casing **900** in the circumferential direction of the rotor **20** in the air gap AG. The image-pickup direction changing part **950** is configured to change the image-pickup direction of the camera **910** to the inclined angle to the radial direction of the rotor **20** in the air gap AG. In this embodiment, this allows the image-pickup position changing part **930** to move the camera **910** to the ventilation hole V provided in the rotor **20**, which makes it easy for the image-pickup direction of the camera **910** to fall into a state along the depth direction of the ventilation hole V, as described above. As a result, the image can be precisely picked up to a deep portion of the ventilation hole V, which enables easy determination of whether or not the ventilation hole V is closed. Accordingly, the inspection apparatus **500** of this embodiment easily enables an efficient inspection.

(133) The inspection apparatus **500** of this embodiment includes the plurality of inspection units **511** to **517** and the coupling shaft **520**. The plurality of inspection units **511** to **517** are provided for performing the inspection on the rotary electric machine **10** in the air gap AG between the rotor **20** and the stator **40**. The coupling shaft **520** is provided for coupling the plurality of inspection units **511** to **517** therebetween. Here, the plurality of inspection units **511** to **517** are configured to be detachable between one another. Further, through each of the plurality of inspection units **511** to **517**, the coupling holes H**510** into which the coupling shafts **520** are inserted in their coupled state are formed. This allows any change in the length of the inspection apparatus **500** according to inspection contents as illustrated in FIG. 3A and FIG. 5 in this embodiment, and efficiently enables

the work of inserting the inspection apparatus **500** into the air gap AG between the rotor **20** and the stator **40**. Further, in this embodiment, when a failure occurs in a part of the plurality of inspection units **511** to **517** constituting the inspection apparatus **500**, the failed inspection unit can be replaced.

(134) In the inspection apparatus **500** of this embodiment, the inspection unit **513** includes the frame **630** housing the image-pickup device **631**, in addition to the image-pickup device **631** which is the inspection device. In the frame **630**, at least the surface is formed using resin. In this embodiment, also in each of the frames of the other inspection units **511**, **512**, **514** to **517**, at least the surface is formed using the resin, similarly to the frame **630** of the inspection unit **513**. In this embodiment, this allows a reduction in weight of the inspection apparatus **500**, which makes it possible to prevent damage such as a flaw to an inspection object from occurring even though the inspection apparatus **500** is in contact with the inspection object. Further, the number of assembly parts such as screws can be reduced, which allows a reduction of the risk due to a fall-off of the assembly parts such as screws.

(135) In the inspection apparatus **500** of this embodiment, the frame **630** of the inspection unit **513** is in the arc shape along the shape of the air gap AG between the rotor **20** and the stator **40**. In this embodiment, the frames of the other inspection units **511**, **512**, **514** to **517** are also each in the arc shape along the shape of the air gap AG similarly to the frame **630** of the inspection unit **513**. In this embodiment, this allows easy insertion of the inspection apparatus **500** into the air gap AG in the cylindrical shape, and allows smooth movement of the inspection apparatus **500** in the air gap AG.

[G] Modified Example

(136) A modified example of the above-described embodiment will be described.

(137) FIG. **11** is a perspective view illustrating the inspection unit **513** in the modified example of the embodiment.

(138) As illustrated in FIG. **11**, in the inspection unit **513**, a guide hole **A61** is formed in the frame **630**. The guide hole **A61** is formed in the first frame member **630a** of the frame **630** so as to extend along a direction in which the image-pickup device **631** moves.

(139) Further, a pair of the pins **970** provided for the image-pickup device **631** is inserted inside the guide hole **A61**. When the image-pickup device **631** moves in the circumferential direction of the rotor **20** (the arrow **AR1**), a pair of the pins **970** swings inside the guide hole **A61**. In this modified example, this causes the image-pickup device **631** to move along the guide hole **A61**, which enables smooth movement of the image-pickup device **631**.

(140) Further, as a matter of course, the inspection apparatus **500** of the above-described embodiment may be configured to enable various kinds of inspection in addition to the above-described inspection. For example, the inspection apparatus **500** may be configured to further include an inspection unit for conducting an electromagnetic core imperfection detection (EL-CID) test on the stator **40**.

OTHERS

(141) While certain embodiments of the present invention have been described above, these embodiments have been presented by way of example only, and are not intended to limit the scope of the inventions. Indeed, these embodiments described herein may be embodied in a variety of other forms; furthermore, various omissions, substitutions, and changes in the form of the embodiments described herein may be made without departing from the spirit of the inventions. The accompanying claims and their equivalents are intended to cover such forms or modifications as would fall within the scope and spirit of the invention.

REFERENCE SIGNS LIST

(142) **10**: rotary electric machine, **20**: rotor, **40**: stator, **41**: stator core, **42**: stator coil, **60**: rotary electric machine casing, **61**: inner casing, **62**: outer casing, **80**: gas cooler, **90**: fan, **200**: rotor core, **201**: rotary shaft, **500**: inspection apparatus, **511**: inspection unit, **511M**: male fitting portion, **512**:

inspection unit, **512F**: female fitting portion, **512M**: male fitting portion, **513**: inspection unit (rotor image-pickup unit), **513F**: female fitting portion, **513M**: male fitting portion, **514**: inspection unit, **514F**: female fitting portion, **515**: inspection unit, **516**: inspection unit, **517**: inspection unit, **600**: control apparatus, **611**: image-pickup device, **612**: lighting device, **621**: guide plate, **630**: frame, **630a**: first frame member, **630b**: second frame member, **631**: image-pickup device (rotor image-pickup device), **632**: flexible cable, **641**: crawler, **642**: permanent magnet, **643**: hammer, **644**: microphone, **645**: image-pickup device, **661**: guide plate, **671**: image-pickup device, **700**: cable, **800**: inspection system, **801**: rack gear, **900**: carriage casing, **910**: camera, **910a**: image-pickup lens, **910b**: lighting section, **911**: camera holder, **911a**: shaft pin, **912**: spur gear, **913**: bush, **914**: bush, **930**: image-pickup position changing part, **950**: image-pickup direction changing part, **AG**: air gap, **AX**: rotation axis, **CG**: cooling gas, **HS**: hexagon socket set screw, **H510**: coupling hole, **K61**: through hole, **K611**: opening, **K62**: through hole, **SC1**: screw, **SC2**: screw, **SP910**: housing space, **SP930**: housing space, **SP950**: housing space

Claims

1. An inspection apparatus which, when an inspection is performed on a rotary electric machine in which an air gap is interposed between a rotor and a stator, and a ventilation hole along a radial direction of the rotor is formed in the rotor, is inserted into the air gap, and adheres to the stator, and travels along an axial direction of the rotor, the inspection apparatus comprising: a rotor image-pickup unit for picking up an image of the rotor using a rotor image-pickup device in the air gap, wherein the rotor image-pickup device comprises: a carriage casing; a camera installed in the carriage casing, and configured to pick up an image of the ventilation hole of the rotor; an image-pickup position changing part installed in the carriage casing, and configured to change an image-pickup position of the camera by moving the carriage casing in a circumferential direction of the rotor; and an image-pickup direction changing part installed in the carriage casing, and configured to change an image-pickup direction of the camera to an inclined angle to the radial direction of the rotor, with the axial direction serving as a support axis.
2. The inspection apparatus according to claim 1, wherein: the rotor image-pickup unit includes a frame housing the rotor image-pickup device, wherein the frame includes a rack gear; and the image-pickup position changing part includes: an image-pickup position changing motor; and an image-pickup position changing pinion gear installed on a rotation shaft of the image-pickup position changing motor, and configured to engage with the rack gear, wherein the rotor image-pickup device is configured to move inside the frame by the image-pickup position changing motor rotating the image-pickup position changing pinion gear.
3. The inspection apparatus according to claim 2, wherein the frame is in an arc shape along a shape of the air gap.
4. The inspection apparatus according to claim 1, wherein: the rotor image-pickup device comprises a camera holder installed in the carriage casing to swing at the inclined angle to the radial direction of the rotor, and housing the camera, wherein the camera holder includes a spur gear; and the image-pickup direction changing part includes: an image-pickup direction changing motor; and an image-pickup direction changing pinion gear installed on a rotation shaft of the image-pickup direction changing motor, and configured to engage with the spur gear, wherein an image-pickup direction of the camera is changed by the image-pickup direction changing motor rotating the image-pickup direction changing pinion gear.
5. An inspection system comprising: the inspection apparatus according to claim 1; and a control apparatus for controlling operation of the inspection apparatus based on an operation command.
6. An inspection apparatus which, when an inspection is performed on a rotary electric machine in which an air gap is interposed between a rotor and a stator, is inserted into the air gap, and travels along an axial direction of the rotor, the inspection apparatus comprising: a plurality of inspection

units for performing the inspection in the air gap; and a coupling shaft for coupling the plurality of inspection units therebetween, wherein: the plurality of inspection units are configured to be detachable between one another in the axial direction; and through each of the plurality of inspection units, a coupling hole into which the coupling shaft is inserted in a coupled state is formed.

7. The inspection apparatus according to claim 6, wherein: each of the plurality of inspection units includes: a frame; and an inspection device provided in the frame; and in the frame, at least a surface is formed using resin.

8. The inspection apparatus according to claim 7, wherein the frame is in an arc shape along a shape of the air gap.
